
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven review of how vaccines transformed public health, 1796–2020

DATA SOURCE

Our World in Data

DATE

April 2026



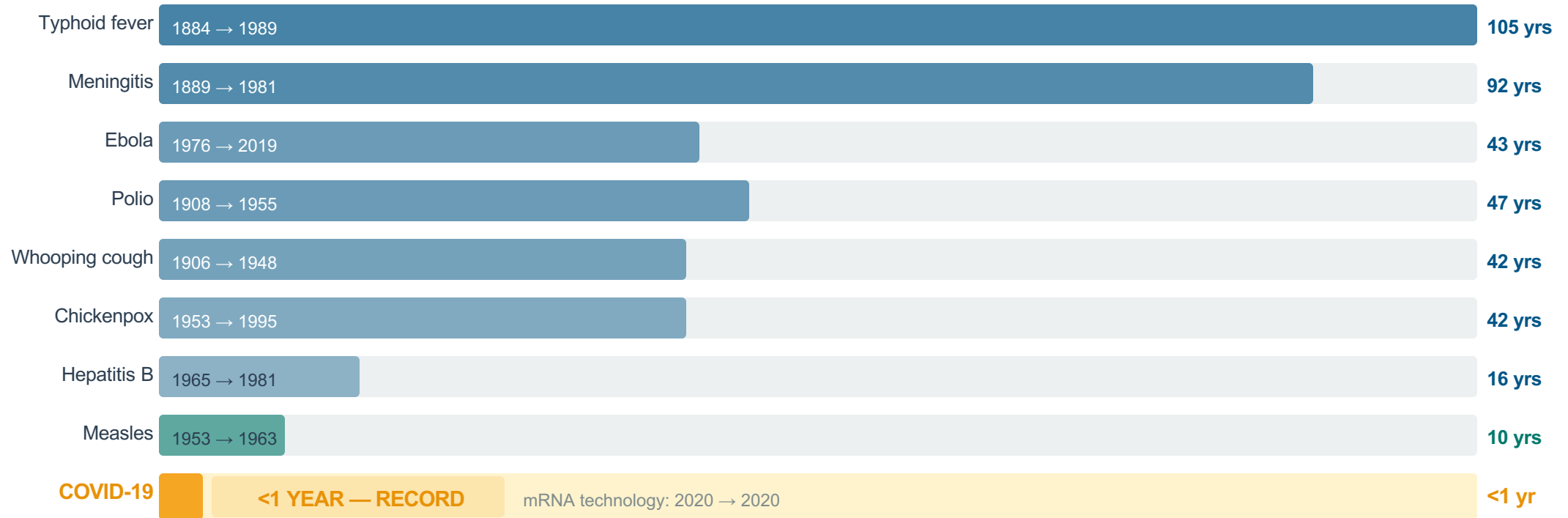
PUBLIC HEALTH

KEY INSIGHT

mRNA technology and genomic tools compressed vaccine development from over a century to less than one year

Years from Pathogen Identification to Vaccine Licensure

Sorted by duration (longest to shortest)



BREAKTHROUGH:

Advanced microscopy, genome sequencing, and mRNA enabled record speed

0

25 yrs

50 yrs

75 yrs

105 yrs

03

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease Burden Reduction

KEY INSIGHT Vaccines achieved complete elimination (100% case and death reduction) for diphtheria, smallpox, and acute polio in the United States

DIPHTHERIA

100%

Cases: 158 → 0 /M/yr



SMALLPOX

100%

Cases: 250 → 0 /M/yr



ACUTE POLIO

100%

Cases: 141 → 0 /M/yr



MEASLES DEATHS

100%

Deaths: 2.5 → 0 /M/yr



Disease	Pre-Vaccine Cases/M/yr	Post-Vaccine Cases/M/yr	Case Reduction	Death Reduction	Status
Diphtheria	158	0	100%	100%	ELIMINATED
Smallpox	250	0	100%	100%	ELIMINATED
Acute polio	141	0	100%	100%	ELIMINATED
Measles	3,044	0.2	99.99%	100%	Near-zero
Rubella	242	0.04	99.98%	100%	Near-zero
Pertussis	1,534	52	96.6%	99.7%	Residual burden
Hepatitis A	465	51	89%	88.7%	Residual burden
Acute Hepatitis B	273	44	83.9%	83.6%	Incomplete
Pneumococcal	233	139	40.5%	31.3%	Major gap

Source: Roush SW and Murphy TV (2007), JAMA. Data normalized to cases/deaths per million population per year.

KEY INSIGHT

Pre-1963: 3,044 cases/million/yr across nearly all states; Post-1980 mandate: near zero — only sporadic outbreaks remained

**1963 First Measles Vaccine**

John Enders develops first vaccine
Pre-1963: nearly all states reported
100–1,000+ cases per 100,000

Cases: 3,044 /million/yr

1971 MMR Vaccine Introduced

Maurice Hilleman combines measles,
mumps, and rubella vaccines

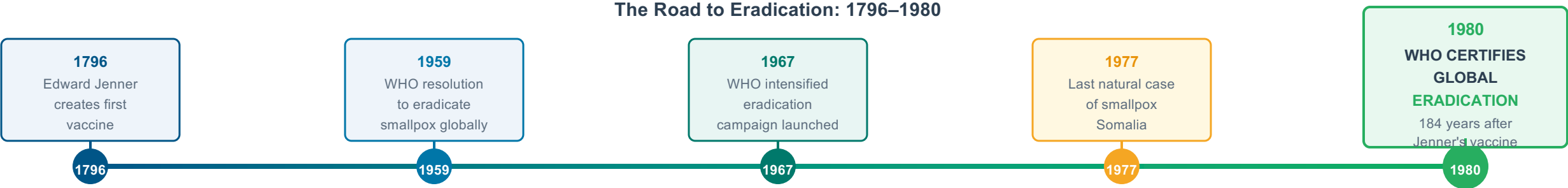
Incidence declining but still significant

1980 Kindergarten Mandate

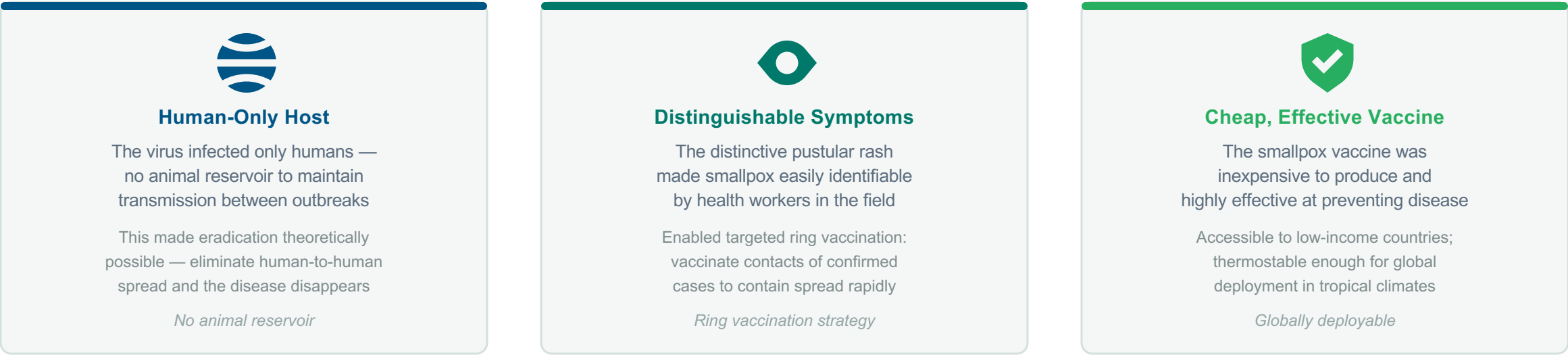
Vaccination required for all children
entering kindergarten

Result: Cases dropped to near zero
across all 50 US states

HISTORIC ACHIEVEMENT Smallpox was declared eradicated in 1980 — the only human disease ever eliminated through a global vaccination campaign



Why Smallpox Was Uniquely Suited for Eradication



KEY INSIGHT

When enough people are vaccinated, pathogens cannot spread — shielding those who cannot be vaccinated at all

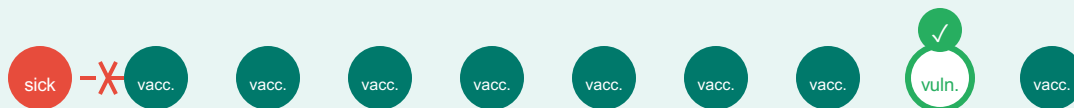
Scenario A: Low Vaccination Coverage — Disease Spreads



~25% vaccinated — pathogen spreads freely — vulnerable individuals infected

No herd immunity established

Scenario B: High Vaccination Coverage — Herd Immunity Achieved



~90% vaccinated — transmission chain broken — vulnerable individuals PROTECTED

Herd immunity shields those who cannot be vaccinated

The Herd Immunity Mechanism

When a sufficient proportion of a population is immune, pathogens cannot find enough susceptible hosts to sustain transmission. Transmission chains break before reaching those who cannot be protected directly.

Who Benefits from Herd Immunity?



Newborns

Infants too young to receive vaccines rely entirely on community immunity for protection in their first months of life



Immunocompromised Individuals

Cannot mount an effective immune response — vaccines may not work or may be unsafe
HIV patients, transplant recipients, and others



Cancer Patients on Chemotherapy

Treatment suppresses the immune system — vaccines are medically contraindicated
Community immunity is their only protection

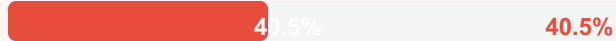
CRITICAL GAP Despite remarkable progress, key diseases remain incompletely controlled — access, supply, and trust are systemic barriers



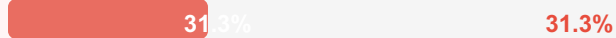
Pneumococcal

Disease

Case reduction:



Death reduction:



Pre-vaccine: 233 cases/M/yr
Post-vaccine: 139 cases/M/yr

Major gap — needs new vaccines + access



Hepatitis A

Vaccine-preventable

Case reduction:



Death reduction:



Pre-vaccine: 465 cases/M/yr
Post-vaccine: 51 cases/M/yr

Significant progress — gap remains
Systemic Barriers Driving These Gaps



Acute Hepatitis B

Vaccine-preventable

Case reduction:



Death reduction:



Pre-vaccine: 273 cases/M/yr
Post-vaccine: 44 cases/M/yr

Significant progress — gap remains



Access Gaps

Not everyone globally has access to vaccines; low-income countries face persistent inequities in supply
Equitable distribution is a policy priority



Supply Shortages

Vaccine supply shortages put children at risk during critical immunization windows in their first years of life
Manufacturing and distribution resilience needed



Trust & Misinformation

Poor scientific understanding can reduce trust and uptake — leaving communities below herd immunity
Science communication and community engagement



100% Elimination Achieved

Diphtheria, smallpox, and acute polio are completely eliminated in the US

Measles cases reduced by 99.99% (3,044 → 0.2 /M/yr)

Rubella cases reduced by 99.98% (242 → 0.04 /M/yr)

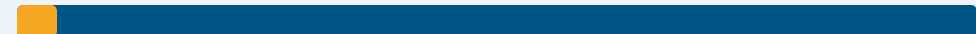
Smallpox: only human disease eradicated worldwide (1980)

→ **Definitive proof that vaccines can eliminate ancient killers**



Vaccine Development Accelerating

From 105 years (typhoid) to under 1 year (COVID-19) — a 100x compression



1884 COVID <1yr

2020

Enabled by mRNA technology, genome sequencing, advanced microscopy, and virus-like particles



Herd Immunity Protects Communities

Population-level vaccination shields newborns, immunocompromised, and cancer patients

When enough people are vaccinated, pathogens cannot spread — protecting those who cannot receive vaccines themselves

→ **Vaccination is a community obligation, not just individual choice**



Coverage Gaps Persist

Pneumococcal: only 40.5% case reduction

Hepatitis A: 89% | Hepatitis B: 83.9%

Access gaps, supply shortages, and trust barriers prevent full disease control globally

Newer, more effective vaccines and improved delivery systems are urgently needed

→ **Urgent: improve access, coverage, and trust**



CALL TO ACTION FOR GLOBAL HEALTH SYSTEMS:

Strengthen access, expand coverage, and rebuild trust — the tools exist; the imperative is political and operational.